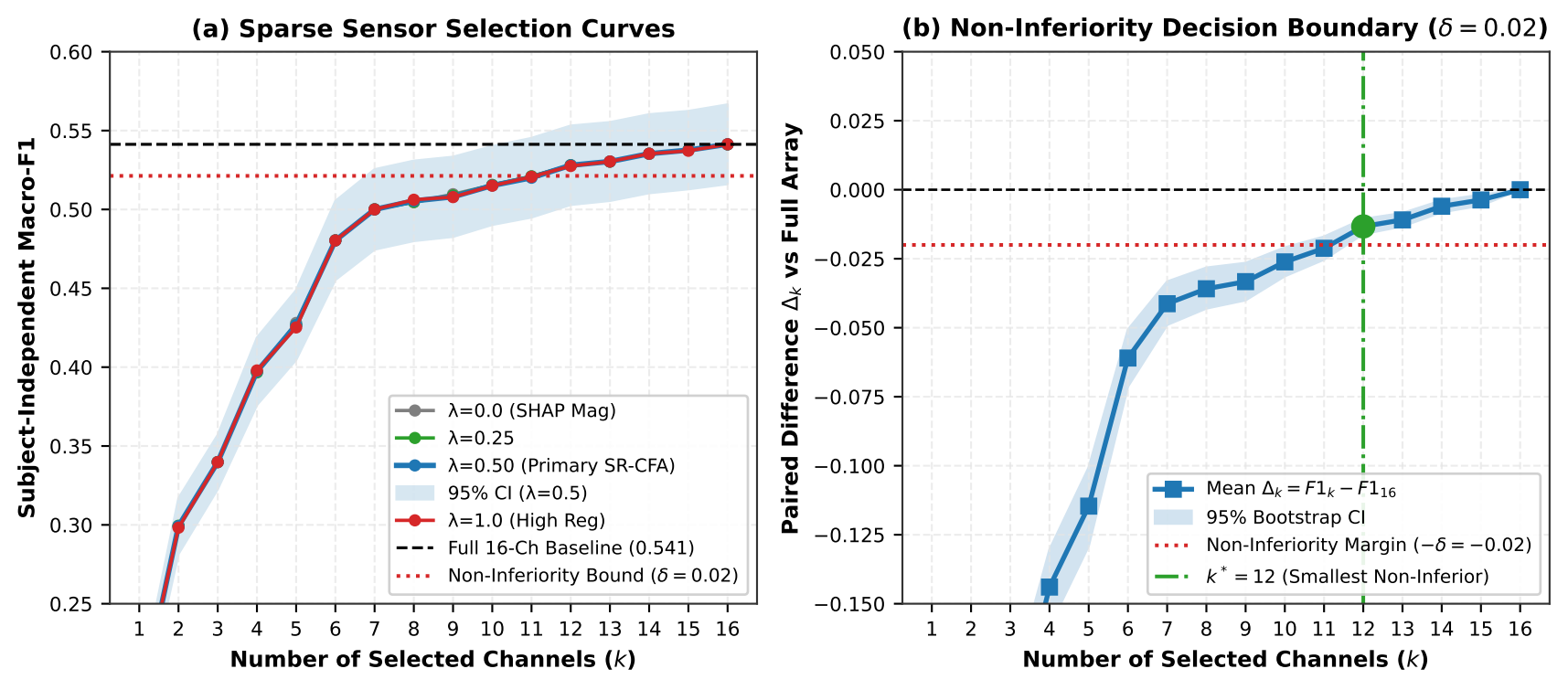
(a) Sparse Sensor Selection Curves
Subject-Independent Macro-F1
Number of Selected Channels (k)
λ=0.0 (SHAP Mag)
λ=0.25
λ=0.50 (Primary SR-CFA)
95% CI (λ=0.5)
λ=1.0 (High Reg)
Full 16-Ch Baseline (0.541)
Non-Inferiority Bound (δ = 0.02)
(b) Non-Inferiority Decision Boundary (δ = 0.02)
Paired Difference Δk vs Full Array
Number of Selected Channels (k)
Mean Δk = F1k − F116
95% Bootstrap CI
Non-Inferiority Margin (−δ = −0.02)
k* = 12 (Smallest Non-Inferior)